FairCKD: Fairness Audit Report

Investigating Bias and Fairness in Machine Learning Models for Automated CKD Diagnosis
M.Tech Dissertation Phase-II · Jimil Jingar · PIET, Parul University

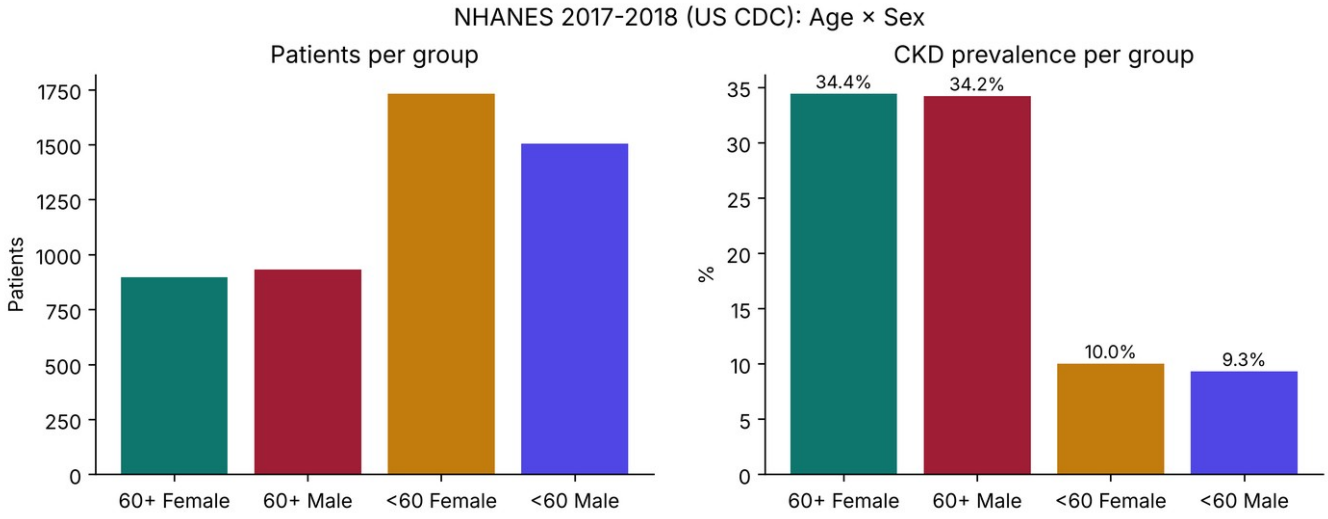
	Value
Dataset	NHANES 2017-2018 (US CDC)
Feature set	Routine blood tests (17 features, no kidney-function tests)
Sensitive attribute	Age × Sex
Patients	5063
Features	17
Groups	60+ Female, 60+ Male, <60 Female, <60 Male
Decision threshold	training-set CKD prevalence
CV protocol	stratified 5-fold, 5 repeats (baseline), 5 repeats (mitigation)

Key findings

• Best baseline model: **Random Forest** (accuracy 0.741, balanced accuracy 0.738, AUC 0.808).
• Bias detection: in **7 of 7** models, CKD is detected least often in the **<60 Female** group (Equal Opportunity gap 0.451–0.626).
• Even a perfect classifier has DP diff = 0.251 and DI = 0.271 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **Unawareness**, Equalized Odds gap 0.619 → 0.530, balanced accuracy 0.736 → 0.728.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **Unawareness**, Equalized Odds gap 0.578 → 0.500, balanced accuracy 0.737 → 0.724.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.623 → **0.425** compared with accuracy-driven model selection, balanced accuracy 0.732 → 0.711.
• Statistical significance: 10 of 10 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, $p < 0.05$).
• External validation (NHANES 2015–16): baseline balanced accuracy 0.740–0.749, Equalized Odds gap 0.488–0.560; best mitigated gap 0.097.

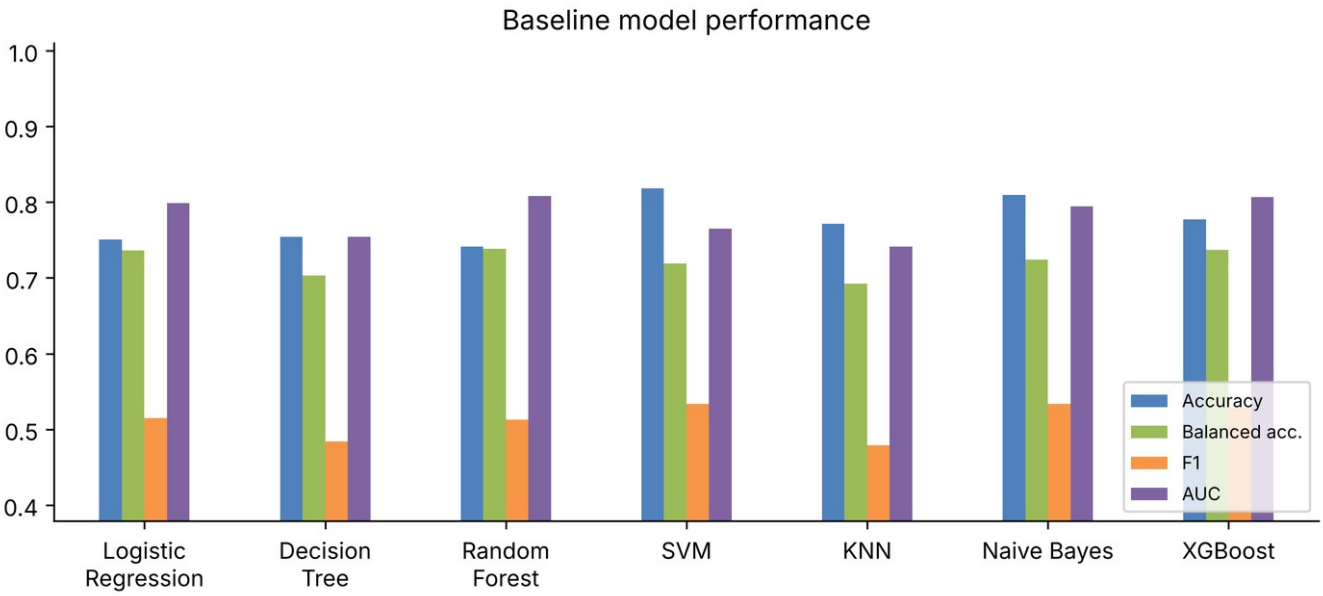
1. Dataset and groups

Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
60+ Female	896	17.7	308	34.4	0.170
60+ Male	931	18.4	318	34.2	0.100
<60 Female	1731	34.2	173	10	0.180
<60 Male	1505	29.7	140	9.3	0.140
All	5063	100	939	18.5	0.150


NHANES 2017-2018 (US CDC): Age × Sex
Patients per group
Patients
0
250
500
750
1000
1250
1500
1750
60+ Female
60+ Male
<60 Female
<60 Male
CKD prevalence per group
%
0
5
10
15
20
25
30
35
34.4%
34.2%
10.0%
9.3%
60+ Female
60+ Male
<60 Female
<60 Male

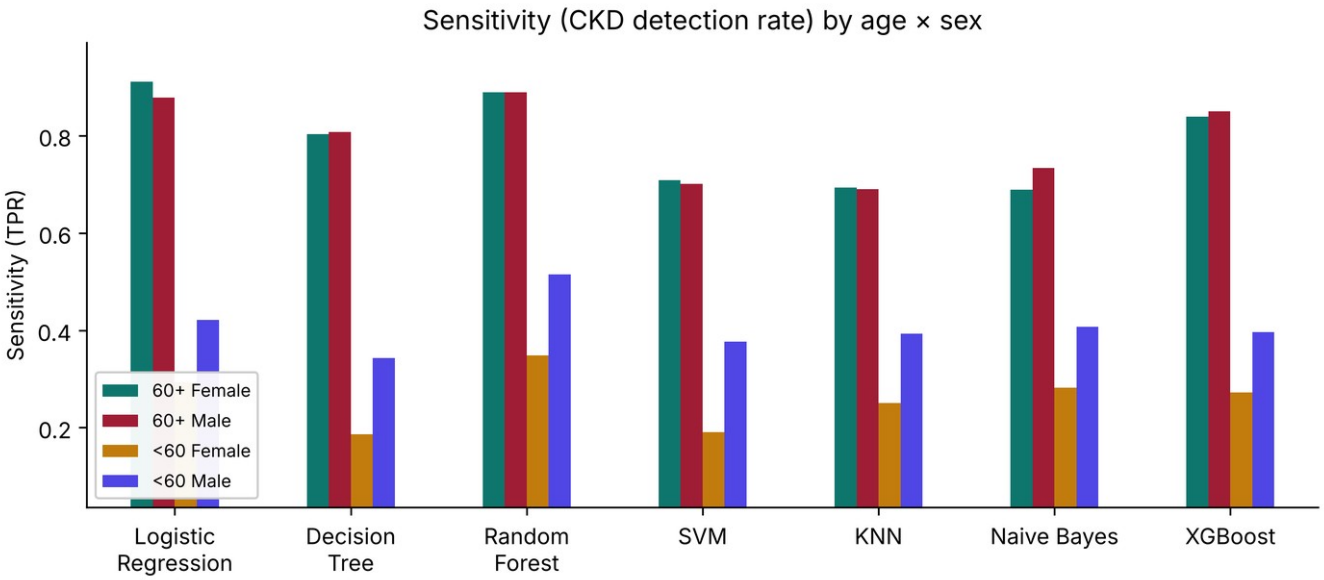
2. Baseline model performance

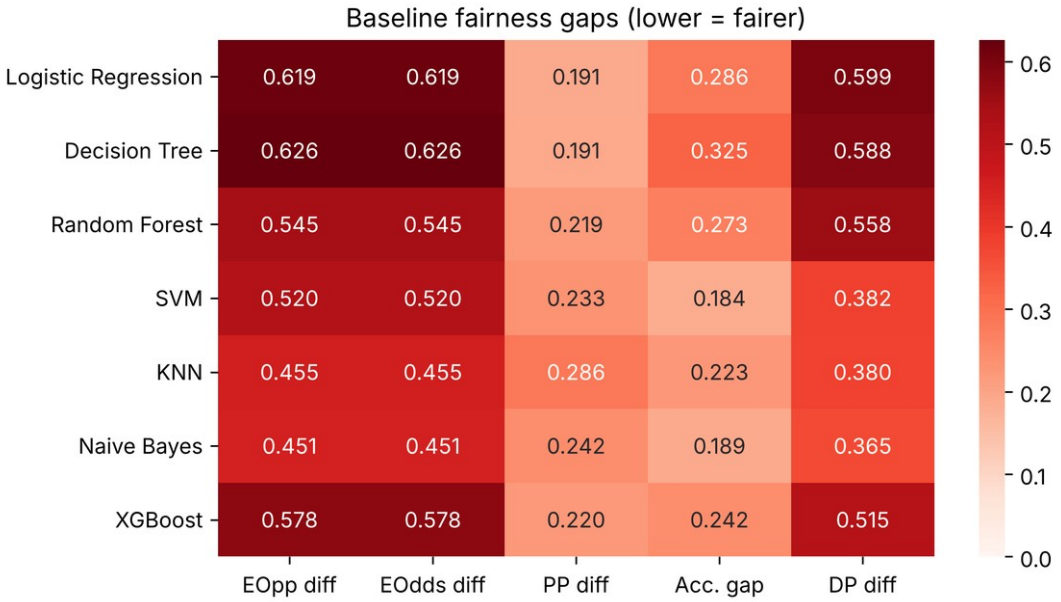
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.751 ± 0.000	0.403 ± 0.000	0.713 ± 0.001	0.515 ± 0.001	0.799 ± 0.001	0.736 ± 0.001
Decision Tree	0.754 ± 0.012	0.397 ± 0.012	0.622 ± 0.030	0.484 ± 0.004	0.754 ± 0.004	0.703 ± 0.005
Random Forest	0.741 ± 0.003	0.394 ± 0.003	0.734 ± 0.002	0.513 ± 0.003	0.808 ± 0.002	0.738 ± 0.002
SVM	0.818 ± 0.002	0.509 ± 0.005	0.561 ± 0.003	0.534 ± 0.004	0.765 ± 0.003	0.719 ± 0.002
KNN	0.772 ± 0.002	0.415 ± 0.004	0.566 ± 0.007	0.479 ± 0.005	0.741 ± 0.006	0.692 ± 0.004
Naive Bayes	0.810 ± 0.002	0.489 ± 0.004	0.587 ± 0.002	0.534 ± 0.003	0.795 ± 0.001	0.724 ± 0.002
XGBoost	0.777 ± 0.003	0.435 ± 0.004	0.672 ± 0.009	0.528 ± 0.005	0.806 ± 0.003	0.737 ± 0.004



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.599 ± 0.005	0.161 ± 0.004	0.619 ± 0.014	0.619 ± 0.014	0.191 ± 0.008	0.286 ± 0.006
Decision Tree	0.588 ± 0.053	0.116 ± 0.019	0.626 ± 0.042	0.626 ± 0.042	0.191 ± 0.032	0.325 ± 0.031
Random Forest	0.558 ± 0.006	0.206 ± 0.004	0.545 ± 0.013	0.545 ± 0.013	0.219 ± 0.008	0.273 ± 0.007
SVM	0.382 ± 0.002	0.132 ± 0.006	0.520 ± 0.013	0.520 ± 0.013	0.233 ± 0.018	0.184 ± 0.003
KNN	0.380 ± 0.005	0.235 ± 0.005	0.455 ± 0.018	0.455 ± 0.018	0.286 ± 0.019	0.223 ± 0.007
Naive Bayes	0.365 ± 0.002	0.202 ± 0.006	0.451 ± 0.011	0.451 ± 0.011	0.242 ± 0.009	0.189 ± 0.003
XGBoost	0.515 ± 0.006	0.164 ± 0.009	0.578 ± 0.025	0.578 ± 0.025	0.220 ± 0.014	0.242 ± 0.005
Perfect classifier (reference)	0.251	0.271	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by age × sex
Sensitivity (TPR)
0.8
0.6
0.4
0.2
60+ Female
60+ Male
<60 Female
<60 Male
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.619 0.619 0.191 0.286 0.599
Decision Tree 0.626 0.626 0.191 0.325 0.588
Random Forest 0.545 0.545 0.219 0.273 0.558
SVM 0.520 0.520 0.233 0.184 0.382
KNN 0.455 0.455 0.286 0.223 0.380
Naive Bayes 0.451 0.451 0.242 0.189 0.365
XGBoost 0.578 0.578 0.220 0.242 0.515
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.6
0.5
0.4
0.3
0.2
0.1
0.0

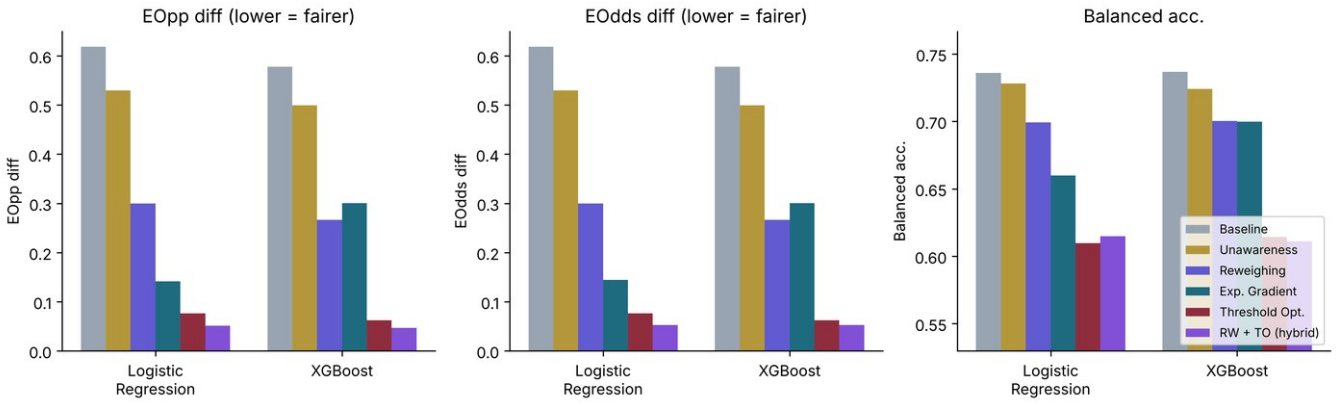
4. Bias mitigation

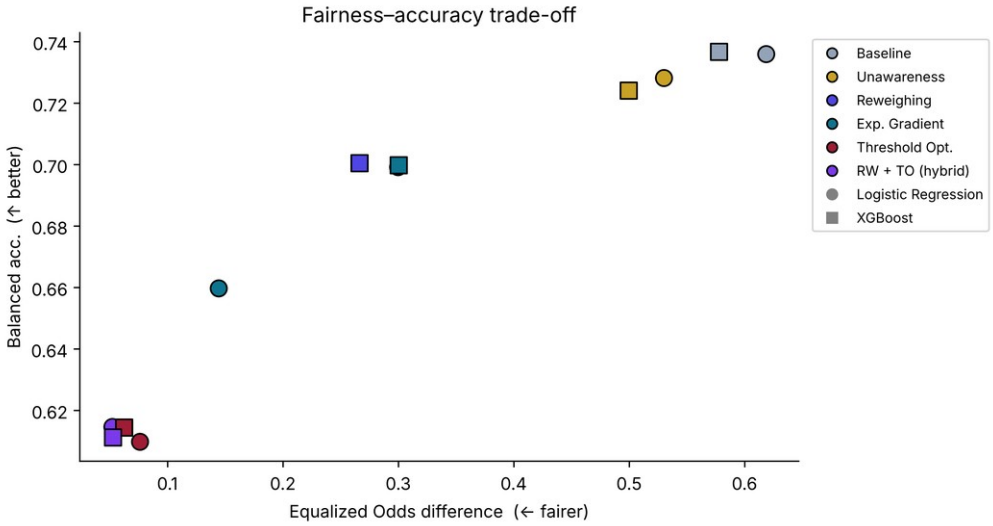
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.751 ± 0.000	0.736 ± 0.001	0.713 ± 0.001	0.515 ± 0.001
Logistic Regression	Unawareness	0.759 ± 0.001	0.728 ± 0.001	0.678 ± 0.002	0.511 ± 0.001
Logistic Regression	Reweighing	0.723 ± 0.002	0.699 ± 0.002	0.661 ± 0.004	0.470 ± 0.003
Logistic Regression	Exp. Gradient	0.694 ± 0.007	0.660 ± 0.006	0.605 ± 0.006	0.423 ± 0.007
Logistic Regression	Threshold Opt.	0.695 ± 0.018	0.610 ± 0.007	0.475 ± 0.025	0.366 ± 0.010
Logistic Regression	RW + TO (hybrid)	0.724 ± 0.009	0.615 ± 0.007	0.441 ± 0.012	0.372 ± 0.010
XGBoost	Baseline	0.777 ± 0.003	0.737 ± 0.004	0.672 ± 0.009	0.528 ± 0.005
XGBoost	Unawareness	0.770 ± 0.002	0.724 ± 0.004	0.651 ± 0.010	0.512 ± 0.005
XGBoost	Reweighing	0.755 ± 0.002	0.700 ± 0.005	0.614 ± 0.008	0.481 ± 0.006
XGBoost	Exp. Gradient	0.785 ± 0.003	0.700 ± 0.005	0.564 ± 0.009	0.493 ± 0.007
XGBoost	Threshold Opt.	0.679 ± 0.042	0.614 ± 0.012	0.511 ± 0.047	0.372 ± 0.014
XGBoost	RW + TO (hybrid)	0.703 ± 0.013	0.611 ± 0.011	0.465 ± 0.030	0.367 ± 0.014

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.619 ± 0.014	0.619 ± 0.014	0.191 ± 0.008	0.599 ± 0.005	0.161 ± 0.004	0.286 ± 0.006
Logistic Regression	Unawareness	0.530 ± 0.005	0.530 ± 0.005	0.229 ± 0.003	0.485 ± 0.002	0.195 ± 0.003	0.218 ± 0.002
Logistic Regression	Reweighing	0.300 ± 0.004	0.300 ± 0.004	0.335 ± 0.006	0.289 ± 0.007	0.458 ± 0.010	0.117 ± 0.009
Logistic Regression	Exp. Gradient	0.142 ± 0.038	0.144 ± 0.035	0.392 ± 0.014	0.171 ± 0.045	0.618 ± 0.082	0.081 ± 0.029
Logistic Regression	Threshold Opt.	0.076 ± 0.035	0.076 ± 0.035	0.359 ± 0.033	0.068 ± 0.009	0.798 ± 0.036	0.066 ± 0.012
Logistic Regression	RW + TO (hybrid)	0.051 ± 0.019	0.052 ± 0.019	0.384 ± 0.034	0.064 ± 0.010	0.782 ± 0.028	0.090 ± 0.004
XGBoost	Baseline	0.578 ± 0.025	0.578 ± 0.025	0.220 ± 0.014	0.515 ± 0.006	0.164 ± 0.009	0.242 ± 0.005
XGBoost	Unawareness	0.500 ± 0.019	0.500 ± 0.019	0.265 ± 0.004	0.429 ± 0.004	0.224 ± 0.006	0.185 ± 0.005
XGBoost	Reweighing	0.267 ± 0.018	0.267 ± 0.018	0.406 ± 0.011	0.190 ± 0.008	0.537 ± 0.015	0.068 ± 0.006
XGBoost	Exp. Gradient	0.300 ± 0.017	0.300 ± 0.017	0.395 ± 0.011	0.222 ± 0.011	0.413 ± 0.017	0.107 ± 0.008
XGBoost	Threshold Opt.	0.063 ± 0.023	0.063 ± 0.023	0.342 ± 0.015	0.067 ± 0.012	0.813 ± 0.051	0.060 ± 0.027
XGBoost	RW + TO (hybrid)	0.046 ± 0.030	0.053 ± 0.024	0.354 ± 0.018	0.058 ± 0.009	0.818 ± 0.027	0.074 ± 0.013





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method – baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Unawareness	-0.073	[-0.112, -0.033]	0.000	-0.007	[-0.015, 0.001]	0.070
Logistic Regression	Reweighing	-0.313	[-0.381, -0.249]	0.000	-0.035	[-0.049, -0.021]	0.000
Logistic Regression	Exp. Gradient	-0.480	[-0.551, -0.404]	0.000	-0.082	[-0.100, -0.064]	0.000
Logistic Regression	Threshold Opt.	-0.521	[-0.603, -0.426]	0.000	-0.128	[-0.147, -0.109]	0.000
Logistic Regression	RW + TO (hybrid)	-0.531	[-0.613, -0.427]	0.000	-0.125	[-0.144, -0.105]	0.000
XGBoost	Unawareness	-0.083	[-0.147, -0.024]	0.004	-0.016	[-0.026, -0.005]	0.006
XGBoost	Reweighing	-0.303	[-0.373, -0.227]	0.000	-0.039	[-0.054, -0.024]	0.000
XGBoost	Exp. Gradient	-0.262	[-0.338, -0.193]	0.000	-0.040	[-0.052, -0.026]	0.000
XGBoost	Threshold Opt.	-0.478	[-0.556, -0.388]	0.000	-0.112	[-0.132, -0.092]	0.000
XGBoost	RW + TO (hybrid)	-0.486	[-0.575, -0.383]	0.000	-0.119	[-0.140, -0.099]	0.000

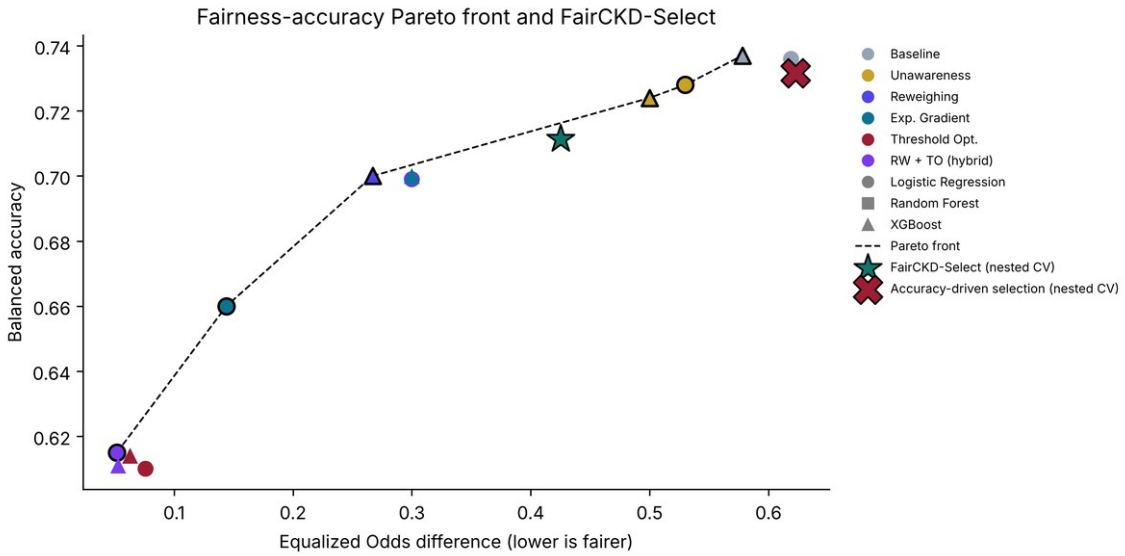
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.767 ± 0.003	0.711 ± 0.005	0.623 ± 0.009	0.425 ± 0.048	0.425 ± 0.048	0.356 ± 0.035
Accuracy-driven selection	0.757 ± 0.003	0.732 ± 0.003	0.691 ± 0.014	0.623 ± 0.010	0.623 ± 0.010	0.567 ± 0.000

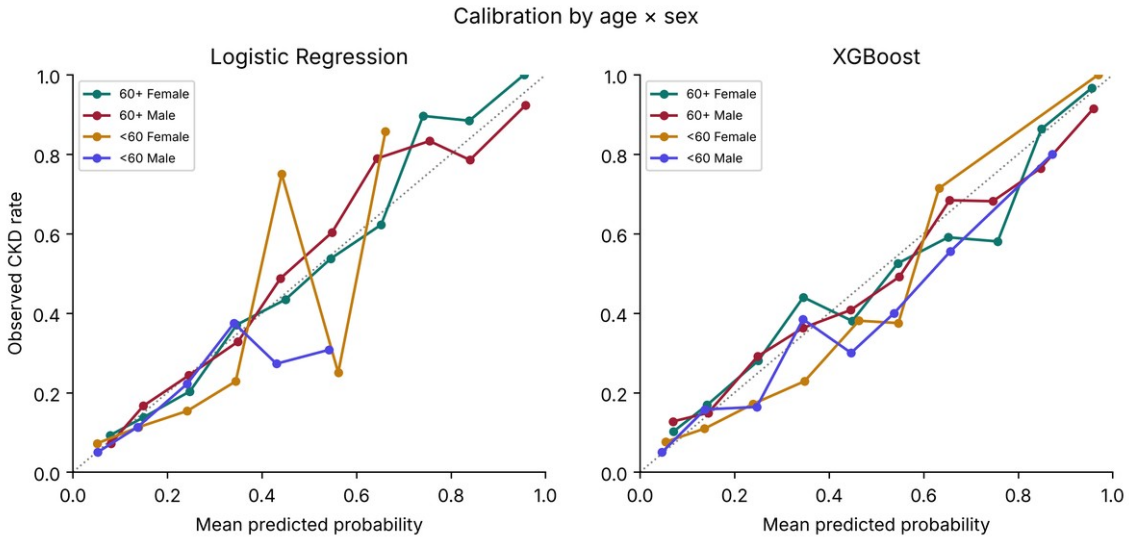
Pipelines chosen by FairCKD-Select across outer folds

Model	Method	Times selected
XGBoost	Baseline	1
XGBoost	Exp. Gradient	2
XGBoost	Reweighing	2
XGBoost	Unawareness	5



4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
60+ Female	896	34.375	34.87	0.495	0.028	0.167	Logistic Regression
60+ Male	931	34.157	32.7	-1.456	0.028	0.171	Logistic Regression
<60 Female	1731	9.994	9.663	-0.331	0.032	0.084	Logistic Regression
<60 Male	1505	9.302	10.184	0.881	0.013	0.074	Logistic Regression
60+ Female	896	34.375	33.331	-1.044	0.042	0.167	XGBoost
60+ Male	931	34.157	33.47	-0.687	0.035	0.172	XGBoost
<60 Female	1731	9.994	9.452	-0.543	0.028	0.084	XGBoost
<60 Male	1505	9.302	9.429	0.127	0.014	0.075	XGBoost

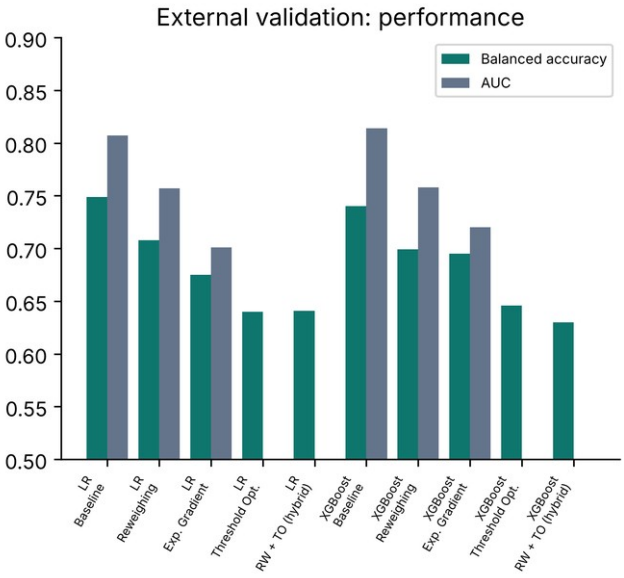


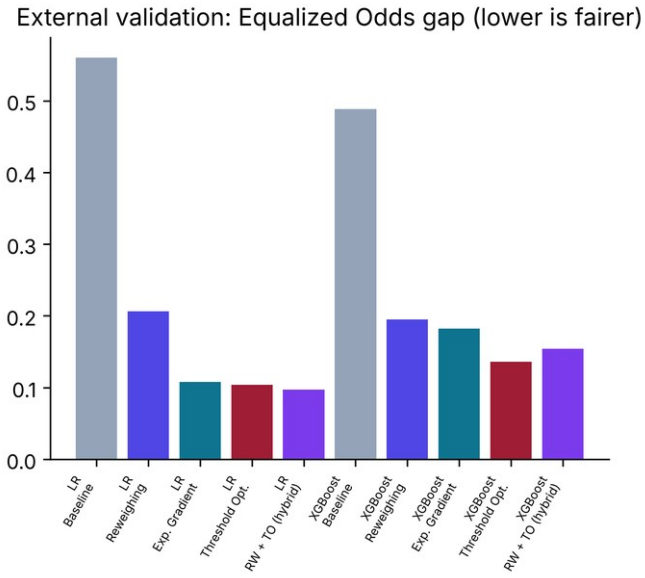
4d. Population-weighted estimates (NHANES survey weights)

Model	W Accuracy	W Balanced acc.	W EOpp diff	W EOdds diff	W DP diff	W TPR [60+ Female]	W TPR [60+ Male]	W TPR [<60 Female]	W TPR [<60 Male]
Logistic Regression	0.788	0.733	0.633	0.633	0.590	0.911	0.904	0.278	0.391
Decision Tree	0.791	0.700	0.627	0.627	0.600	0.821	0.814	0.203	0.318
Random Forest	0.782	0.740	0.604	0.604	0.566	0.905	0.892	0.304	0.513
SVM	0.851	0.705	0.499	0.499	0.351	0.684	0.687	0.195	0.348
KNN	0.806	0.690	0.461	0.461	0.356	0.692	0.660	0.241	0.446
Naive Bayes	0.847	0.712	0.425	0.425	0.334	0.655	0.715	0.290	0.369
XGBoost	0.810	0.724	0.587	0.587	0.522	0.830	0.844	0.258	0.362

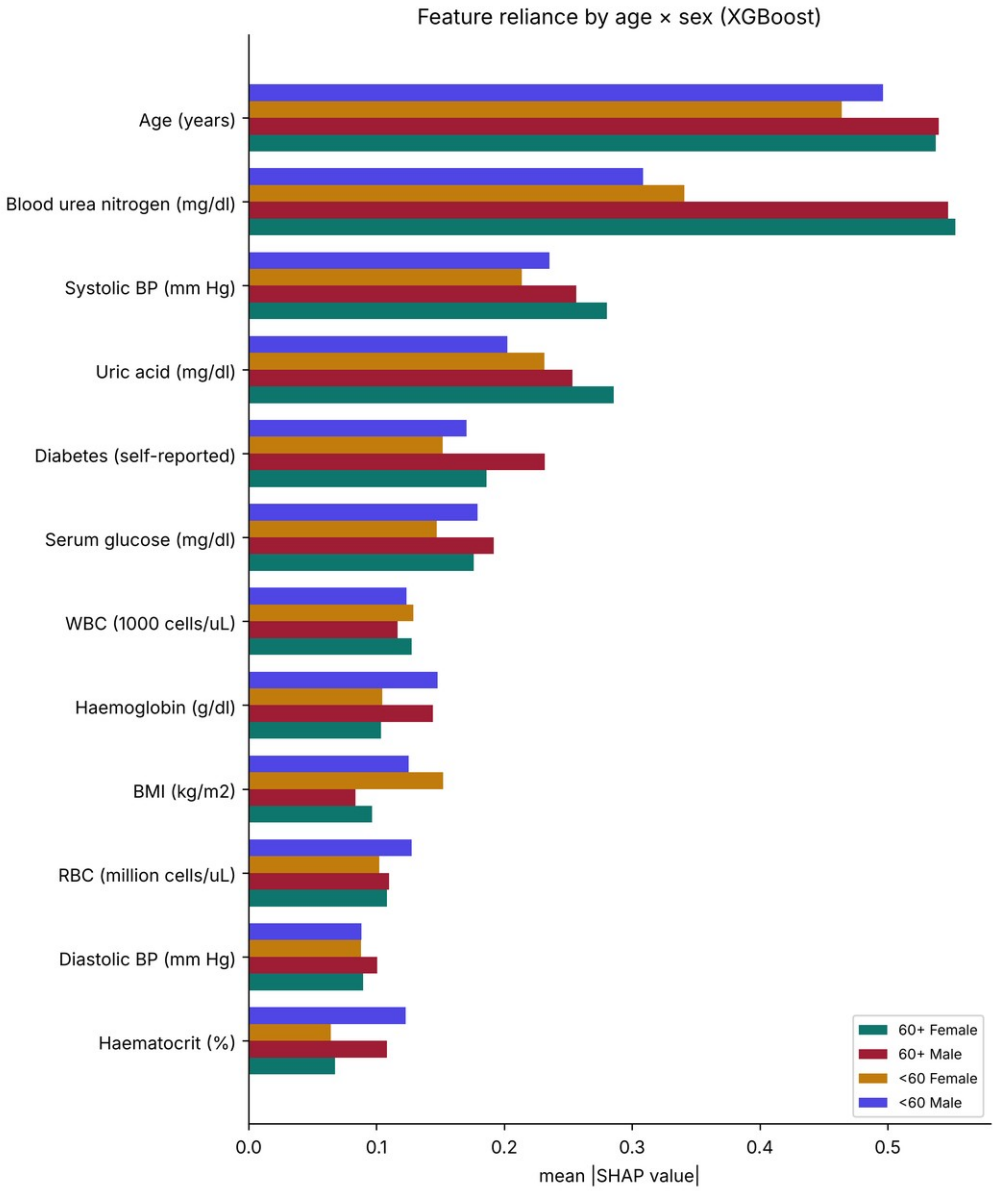
4e. External validation (trained on NHANES 2017–18, tested on NHANES 2015–16)

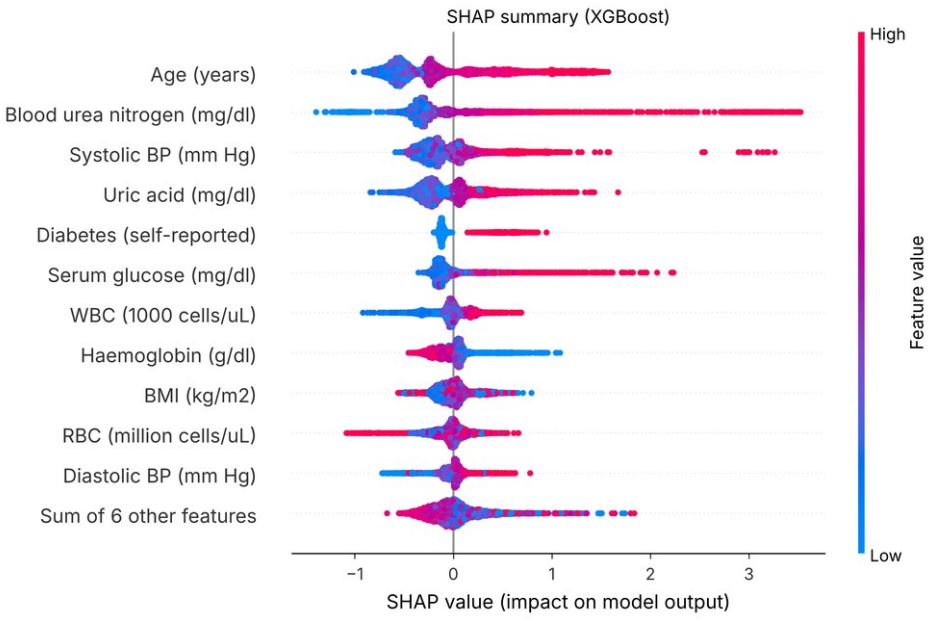
Model	Method	Accuracy	Balanced acc.	Recall	AUC	EOpp diff	EOdds diff
Logistic Regression	Baseline	0.784	0.749	0.696	0.807	0.508	0.560
Logistic Regression	Reweighing	0.744	0.708	0.654	0.757	0.206	0.206
Logistic Regression	Exp. Gradient	0.707	0.675	0.628	0.701	0.108	0.108
Logistic Regression	Threshold Opt.	0.770	0.640	0.447		0.104	0.104
Logistic Regression	RW + TO (hybrid)	0.774	0.641	0.444		0.097	0.097
XGBoost	Baseline	0.791	0.740	0.666	0.814	0.488	0.488
XGBoost	Reweighing	0.750	0.699	0.622	0.758	0.195	0.195
XGBoost	Exp. Gradient	0.788	0.695	0.557	0.720	0.182	0.182
XGBoost	Threshold Opt.	0.677	0.646	0.600		0.136	0.136
XGBoost	RW + TO (hybrid)	0.618	0.630	0.648		0.154	0.154





6. Explainability (SHAP)





Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.